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Review/ Derleme

Synaptic Dysfunction in Alzheimer's Disease: A Narrative Review of Neuronal Changes

Alzheimer Hastalığında Sinaptik İşlev Bozukluğu: Nöronal Değişimlere İlişkin Anlatımsal Bir Derleme

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ABSTRACT

Alzheimer's disease (AD) is a prevalent neurological condition among the elderly, characterized by progressive cognitive decline and is the most common form of dementia. Neurofibrillary tangles (NFTs) and amyloid beta (Aβ) plaques are the hallmark pathological features of AD. However, substantial experimental evidence suggests that alterations in synaptic function, leading to cognitive impairment in AD patients, may represent early indicators of the disease. In this review, we classify, discuss, and describe the primary findings related to changes in the neurophysiological mechanisms of pre- and post-synaptic structure and function underlying AD. Both mechanical and chemical impairments caused by Aβ plaques and NFTs disrupt synaptic function, leading to compromised long-term potentiation (LTP) and long-term depression (LTD), as well as alterations in pre- and post-synaptic proteins.

Keywords: Alzheimer's disease, Pre-synaptic structures, Post-synaptic structures, Synaptic dysfunction

ÖZET

Alzheimer hastalığı (AH), yaşlılar arasında yaygın bir nörolojik durum olup, ilerleyici bilişsel gerileme ile karakterize edilir ve en sık görülen demans türüdür. AH'nin belirgin patolojik özellikleri nörofibriler yumaklar (NFT'ler) ve amiloid beta (Aβ) plaklarıdır. Bununla birlikte, önemli deneysel kanıtlar, AH hastalarında bilişsel bozulmaya yol açan sinaptik fonksiyon değişikliklerinin, hastalığın erken belirtileri olabileceğini göstermektedir. Bu derlemede, AH'nin temelinde yatan pre- ve post-sinaptik yapı ve fonksiyondaki değişikliklerle ilgili ana bulguları sınıflandırıyor, tartışıyor ve açıklıyoruz. Aβ plakları ve NFT'ler tarafından neden olunan mekanik ve kimyasal bozukluklar, sinaptik fonksiyonu bozarak uzun süreli potansiyasyon (USP) ve uzun süreli depresyon (USD) süreçlerini zayıflatmakta ve pre- ve post-sinaptik proteinlerde değişikliklere yol açmaktadır.

Anahtar Kelimeler: Alzheimer hastalığı, Pre-sinaptik yapılar, Post-sinaptik yapılar, Sinaptik işlev bozukluğu

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Introduction

In 1901, Dr. Alois Alzheimer examined Auguste Deter, a 51-year-old patient with severe cognitive decline and behavioral disturbances, including jealousy, disorientation, hallucinations, agnosia, apraxia, aphasia, and aggression (1). Her condition worsened, leading to loss of independence. After her death in 1906, an autopsy revealed neurofibrillary tangles (NFTs), amyloid beta (Aβ) plaques, and dystrophic neurites. In 1907, Alzheimer reported the first case of "presenile dementia," later named Alzheimer's disease (AD) by Emil Kraepelin.

According to the World Health Organization (WHO), dementia affects approximately 55 million individuals worldwide, with over 60% residing in low- and middle-income countries. The projected escalation of this quantity is anticipated to reach 78 million by the year 2030 and surge further to 139 million by 2050 (2), coinciding with the expansion of the elderly demographic across virtually all nations.

The main clinical manifestations of AD include cognitive dysfunction, memory loss, and abnormal changes in personality (3). Two fundamental hypotheses delineate the pathogenesis of AD: the amyloid cascade and tau hyperphosphorylation hypothesis. Pathological hallmarks of AD consist of extracellular Aβ plaques and intracellular NFTs in the brain, accompanied by neuronal and synaptic loss, leading to cognitive impairment and eventual dementia (4). In AD, Aβ plaque accumulation typically begins in neocortical areas and subsequently spreads to other brain regions as the disease advances (5). Similarly, NFTs initially emerge in the medial temporal lobe and later extend toward neocortical regions as the disease progresses (6). However, there is ongoing debate regarding the temporal sequence of these two processes—that is, which occurs first. Some studies suggest that Aβ accumulation precedes NFT formation (7), while others indicate that NFTs may develop prior to Aβ deposition (8).

Synaptic loss and dysfunction, followed by neuronal cell death, are two critical components of the neurodegenerative process in AD, particularly in its early stages (9, 10).

While conventional Alzheimer's research has predominantly concentrated on neuronal death, apoptosis, Aβ plaques, and NFTs at the cellular and molecular levels, the loss of neocortical synapses demonstrates a more robust correlation with cognitive impairment compared to cell count. Furthermore, synapse loss tends to occur earlier and with greater severity than neuronal death (11). Throughout the past three decades, the research has consistently associated synapse loss with compromised cognitive performance. Studies dating back to the early 1990s revealed a reduced cortical synaptic density in individuals with AD (12). Furthermore, contemporary studies have corroborated this finding, indicating that synapse loss constitutes a pivotal pathogenetic factor in AD (13). Synapse loss serves as a direct and early structural indicator of cognitive decline in Alzheimer's patients, exhibiting stronger correlations with cognitive impairment than the extent of Aβ plaques, NFTs, or neuron loss (13, 14).

In addition to synaptic loss, synaptic dysfunction represents an early pathological manifestation and a pivotal component of AD's neurodegenerative process. Epidemiological studies have demonstrated a robust correlation between synaptic loss and cognitive decline in AD patients, suggesting that synaptic integrity plays a causative role in the etiology of AD (15). Research conducted in mouse models further supports the notion that the accumulation of NFTs and Aβ plaques triggers synaptic loss and dysfunction without widespread neuron loss (16). Furthermore, investigations employing electron microscopy or immunohistochemical staining for synaptic markers have revealed comparable reductions in both pre- and post-synaptic components (17). This review explored the impact of Aβ plaques and NFTs on synapse number, structure, and function, as well as the alterations in pre- or post-synaptic dynamics induced by protein aggregation or other contributory factors.

In this review we focused on the presynaptic and postsynaptic changes induced by Aβ plaques and NFTs in AD, and consequently did not address other major contributors, including mitochondrial dysfunction, oxidative stress, alterations in lipid metabolism, and the role of neuroglia; although omitted due to reference constraints, we acknowledge these as important limitations of our review, recognize the critical importance of glial cells in AD-related neurodegeneration and synaptic dysfunction, and emphasize that these mechanisms warrant further detailed investigation.

Following this general overview of AD, we next turn to its defining pathological hallmarks—Aβ plaques and NFTs—which play a pivotal role in disease onset and progression.

Effects of Aβ Plaques and NFTs on Synapses

Aβ Dynamics in AD

The amyloid precursor protein (APP) is an integral membrane glycoprotein expressed in various cell types, including neurons. It consists of extracellular and intracellular domains. APP can undergo sequential proteolytic processing via two pathways: the α pathway (non-amyloidogenic) and the β pathway (amyloidogenic). Typically, APP undergoes sequential cleavage via the α pathway by α-secretase and γ-secretase, resulting in the production of soluble amyloids. However, when APP is sequentially cleaved by β-secretase followed by γ-secretase, β-secretase initially cleaves APP to release a large secreted derivative, soluble peptide APPβ (sAPPβ). Subsequently, a 99-amino acid fragment remains membrane-bound and is rapidly cleaved by γ-secretase to generate Aβ (18).

Despite its ubiquitous expression, APP and its non-amyloid cleavage products have a widespread expression and exert a plethora of physiological effects that profoundly influence neuronal function and neurotransmission(19). These effects encompass crucial roles in neural cell adhesion, neurite and axon outgrowth, neuronal viability, as well as axonal guidance. Investigations employing APP knockout mice have further elucidated changes in the profile of both pre- and post-synaptic proteins, including alterations in postsynaptic density protein 95 (PSD-95) and the α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA) receptor subunit GluR1. This finding indicates that in neurotransmission and plasticity processes, not only the cleavage products of APP but also APP itself plays a role. Notably, APP acts as the progenitor of Aβ peptide, which stands as the predominant constituent of Aβ plaques detected within the brains of individuals suffering from AD (20).

Neurite Dystrophy and Dendritic Spine Density in the Context of Aβ Plaques

In vivo studies have yielded empirical evidence suggesting a correlation between Aβ plaques and neurite dystrophy in animal models. A two-photon microscopy study confirmed findings from postmortem AD patient brains and amyloid mouse models, showing that plaques are associated with dystrophy and alter the trajectory of neurites (21). Many subsequent in vivo imaging investigations have consistently shown the presence of dystrophic neurites around plaques (22, 23).

Dystrophic neurites observed near plaques may suggest either that plaques induce neurite dystrophy or that dystrophic neurites promote plaque development. Longitudinal imaging of dystrophic neurites and plaques indicates that plaque production precedes the appearance of dystrophic neurites (24). Further longitudinal imaging of these areas revealed no evidence of new plaque growth around the dystrophic dendrites. These findings suggest the potential for plaques to augment the prevalence of dystrophic neurites, particularly if the latter do not actively drive plaque formation, given the temporal sequence where dystrophic neurite emergence typically follows plaque development, as elucidated by Blazquez-Llorca et al. (23). Plaques are more likely to increase dystrophic neurites if the latter do not stimulate plaque development. The emergence of dystrophic neurites following plaque formation supports this theory (23). Subsequent to plaque formation, the curvature of neurites around the plaque continues to grow, with damaged neurites appearing in the subsequent days (25). Approximately a quarter of the neurites surrounding plaques exhibit dystrophy, and further research on their dynamics indicates that they are highly stable, with a lifespan of about 76 days (23). The volume of dystrophic neurites is highly variable and surprisingly dynamic, with neurites both shrinking and expanding.

The morphology of dendritic spines, the postsynaptic components of synapses, is strongly associated with learning and memory (26). Notably, alterations in dendritic spines emerge in the early stages of AD, preceding neuronal death and Aβ plaque formation (12).

In addition, dendritic spine density shows a significant decrease within dystrophic neurites proximal to plaques (27, 28). In vivo imaging studies in mouse models of amyloidosis have demonstrated a reduced density of dendritic spines within a 50 μm radius of the plaque compared to non-transgenic controls or regions situated further away from the plaque (29). Furthermore, a reduction in spine density has been documented in dendrites located more than 50 μm from the plaque in transgenic animal models (25). The decrease in spine density with increasing distance from the plaque may be attributed to a lower local concentration of soluble amyloid compared to the vicinity of the plaque. However, reduced spine density does not exhibit a direct correlation with plaque size, suggesting that the plaque itself may not solely be responsible for synapse loss, but rather the soluble amyloid present in the periphery of the plaque (30). A recent study has revealed that significant morphological changes in dendritic spines of hippocampus observed in AD models, including decreased spine density (31).

Aβ Plaques and Synaptic Plasticity in AD

To understand the underlying mechanisms of synaptic dysfunction and neurologic and neuropsychiatric disorders, researchers delve into the realm of synaptic plasticity. Evaluation of synaptic plasticity is a well-known method for understanding the synaptic dysfunction in AD caused by Aβ plaques.

Both strong and weak stimulation induce depolarization of the postsynaptic membrane, leading in the removal of the Mg2+ block from postsynaptic NMDA receptors (NMDARs), enabling substantial or moderate calcium influx into the neuron (32). In healthy synapses, strong Ca^{2+} signals promote long-term potentiation (LTP), while moderate activity activates phosphatases leading to, long-term depression (LTD). In AD, this balance shifts—impairing LTP and facilitating LTD (33).

Alterations in these mechanisms are implicated in nervous system disorders, including AD. Approximately 25 years ago, it was discovered that Aβ rapidly and potently inhibits hippocampal LTP (34) and facilitates LTD (35) at CA3-to-CA1 synapses following injection in rats in vivo. A probable explanation for the impaired plasticity caused by Aβ plaques is the physical obstruction of communication between pre- and post-synaptic compartments.

Another potential mechanism for impaired LTP could involve the failure of inhibition of protein phosphatase 1 (PP1) by protein kinase A (PKA). Knobloch et al. found that inhibiting PP1 in APP transgenic mice restored LTP (36). However, it remains unclear how Aβ prevents the inhibition of PP1. A study utilizing fibrillar aggregates of Aβ demonstrated that Aβ actually inhibits PP1 in vivo, rather than promoting its activity (37).

As previously discussed, the γ-secretase cleavage of the remaining APP fragment generates Aβ, which manifests in various isoforms, each exerting distinct contributions to disease pathology. Pre-fibrillar soluble Aβ oligomers (AβOs) have been shown to bind to the dendritic processes of neurons with high affinity (38). Studies have reported that Aβ and/or AβOs bind to excitatory neurotransmitter receptors, such as NMDA receptor subunits and α7 nicotinic acetylcholine receptors (nAchR), as well as signaling proteins expressed at postsynaptic membranes, such as the prion protein (PrPC) (39).

For an extended period, soluble aggregates of Aβ have been assumed to be the most synaptotoxic products of APP metabolism. However, recent studies have indicated that N-terminally extended Aβ and peptides with a truncated Aβ C-terminus inhibit LTP in vitro (40). The extent to which these synaptotoxic APP metabolites contribute to synaptic disruption in AD remains uncertain. Furthermore, impairments in LTP induced by the synaptic toxicity of soluble AβOs and their binding to synapses require the presence of APP, as demonstrated by Wang et al. (41), suggesting that APP plays a role in AD pathogenesis beyond Aβ production.

In contrast, AβOs have been shown to enhance LTD (35), also mediated by the PrPC (42). LTD has been shown to reduce the size and occurrence of postsynaptic dendritic spines. AβOs have also been associated with impaired LTD. AβOs inhibit glutamate transporters in the postsynaptic membrane, resulting in increased glutamate concentration in the synapse (43). Additionally, excessive glutamate binds to NMDA receptors in the synapse, inducing the influx of Ca2+ ions into the postsynaptic dendrite.

Some studies suggest that in the early stages of AD, Aβ accumulation is the initial trigger of synaptic dysfunction. In other words, Aβ comes first and initiates disruption at the synapses (44).

Tau Protein Dynamics in AD

Tau is a microtubule-associated protein (MAP) that binds to tubulin in a phosphorylation-dependent manner, thereby stabilizing microtubules (MTs) (45). As a phosphoprotein, tau undergoes a cycle of phosphorylation and dephosphorylation, regulated by the balance of protein kinase and protein phosphatase activity. Phosphorylation of tau reduces its affinity for MTs, leading to their destabilization, a process tightly regulated under physiological conditions (46). However, in pathological conditions, tau phosphorylation becomes dysregulated, resulting in hyperphosphorylation, aggregation, and subsequent disruption of MTs (47). The loss of MT-stabilizing properties ultimately induces neuronal dysfunction (48). Aβ has been shown to activate Tau protein kinase 1, leading to abnormal activation and hyperphosphorylation of tau, culminating in the formation of paired helical filaments (PHFs) and NFTs. In neurons, hyperphosphorylated microtubule-associated protein tau undergoes aggregation, ultimately culminating in the formation of NFTs.

Pathophysiological Roles of Tau in AD

NFTs have been associated with a reduction in synaptic density, neurotoxic effects, and cellular dysfunction (49, 50). Experimental evidence indicate a positive correlation between the degree of tau aggregation, the level of tau hyperphosphorylation, and the pathological severity of AD (51). Furthermore, emerging evidence suggests that tau, rather than Aβ, plays a pivotal role in determining cognitive status (52).

The intricate mechanisms underlying tau's preservation of synaptic plasticity or the mechanisms through which its pathogenesis interferes with this process remain incompletely understood. Pathogenic tau may disturb normal synaptic vesicle release at the presynapse. Despite the absence of tau protein itself in the synaptic vesicle proteome of mice (53), hyperphosphorylated tau has been observed to form more stable interactions with synaptic vesicles isolated from AD brains compared to those from normal control brains (54).

The somato-dendritic compartment contains almost the same amount of tau as the axon (55). Synaptic activity directs tau to the postsynaptic density (PSD), where it interacts with postsynaptic density proteins (56). Dendritic tau does not bind to microtubules and is associated with the loss of dendritic spines (57) and is generally hyperphosphorylated (58). Dendritic loss, aberrant postsynaptic activity, and cognitive impairment are all linked to these tau species in AD (59). Dendritic tau has been shown to mediate Aβ-dependent excitotoxicity by complexing with Fyn kinase to stimulate NMDA receptor activation, where dysregulated Fyn–NMDAR coupling subsequently disrupts calcium signaling (60). In tau models, AMPA-mediated currents in dendritic spines are decreased (48). This impairment could be caused by hyperacetylated tau, which reduces KIBRA levels at synapses and impairs AMPA receptor trafficking during plasticity (16). It's still unclear whether hyperphosphorylated tau affects the stability of AMPA receptors at synapses.

Tau is also associated with the endoplasmic reticulum (ER) and the Golgi apparatus in the somatodendritic part of the synapse (61). The physical abnormalities observed in the ER and the Golgi apparatus, along with the atypical N-glycosylation of tau in AD, may be attributed to the hyperphosphorylation of tau and its accumulation in the somatodendritic region (62). In AD, abnormally hyperphosphorylated tau is implicated in granulovacuolar alterations as well as the formation of NFTs (63).

The aggregation of pathogenic tau within synapses intensifies synaptic loss and impairs synaptic plasticity. A robust correlation has been established between NFT density, synaptic loss, and cognitive decline (64). An experimental study using a mouse model of AD has demonstrated that reducing endogenous tau levels effectively mitigates synaptic impairment, likely through alterations in postsynaptic molecules (65). Additionally, synaptic abnormalities have been linked to the soluble form of tau preceding NFT formation, indicating that soluble tau alone can induce synaptic dysfunction (66). Notably, individuals with elevated levels of pathogenic tau exhibit compromised synaptic plasticity and accelerated cognitive decline in AD (67).

Although the precise mechanisms underlying NFT-dependent pathology remain incompletely elucidated, several theories have been proposed. Notably, tau has been identified as a crucial factor in the process of LTD (68). Glycogen synthase kinase-3 (GSK-3), a prominent tau kinase, is intricately involved in NMDA receptor signaling (69), with its activation being requisite for AMPAR internalization. Tau plays a pivotal role in facilitating the interaction between Protein Interacting with C-kinase (PICK1) and GluA2 (GluR2), a molecular event crucial for the internalization of AMPARs. Experimental evidence from biochemical and electrophysiological studies has indicated that site-specific phosphorylation of tau at residue 396 (S396) is not only associated with but also essential for LTD. Furthermore, in the absence of tau, there is a notable reversal of spatial memory deficits, thereby underscoring the pivotal role of tau in hippocampal LTD, as evidenced by Regan et al. (70).

Taken together, in AD, Aβ plaques and NFTs initiate a multi-step cascade in which Aβ and tau pathology lead to synaptic dysfunction and loss, followed by network disintegration and ultimately neuron loss (Figure 1). In the following section, we will turn our focus to synaptic proteins in AD and their relevance to this process.

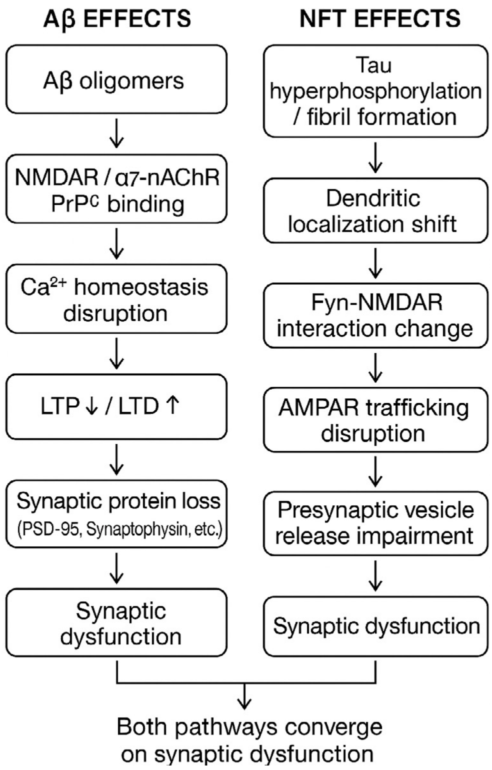

Figure 1 This schematic illustrates the distinct yet converging pathological pathways of amyloid-beta (Aβ) plaques and neurofibrillary tangles (NFTs) in Alzheimer's disease (AD). Aβ oligomers interact with synaptic receptors such as N-methyl-D-aspartate receptors (NMDAR), α7 nicotinic acetylcholine receptor (α7-nAChR), and prion protein (PrPC), triggering calcium dysregulation, long-term potentiation (LTP) impairment, and long-term depression (LTD) enhancement. These events lead to disruptions in synaptic adhesion molecules, such as N-cadherin cleavage, and postsynaptic scaffold proteins, such as Homer1b cluster loss, ultimately contributing to synaptic dysfunction. Hyperphosphorylated tau mislocalizes to dendrites, altering postsynaptic signaling via Fyn–NMDAR coupling changes, and interferes with presynaptic vesicle release and α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA) receptor trafficking, further impairing synaptic transmission. Both Aβ and NFT pathways converge on synaptic dysfunction, which is a core early pathological feature of AD and contributes significantly to cognitive decline.

Synaptic Protein Alterations in AD

Studies partially elucidate the molecular mechanisms at the protein level underlying synaptic dysfunction observed in AD. Alterations in pre- or post-synaptic proteins, situated within synaptic vesicles, the cytoplasm, and terminal membranes, may contribute to the loss of synaptic connectivity.

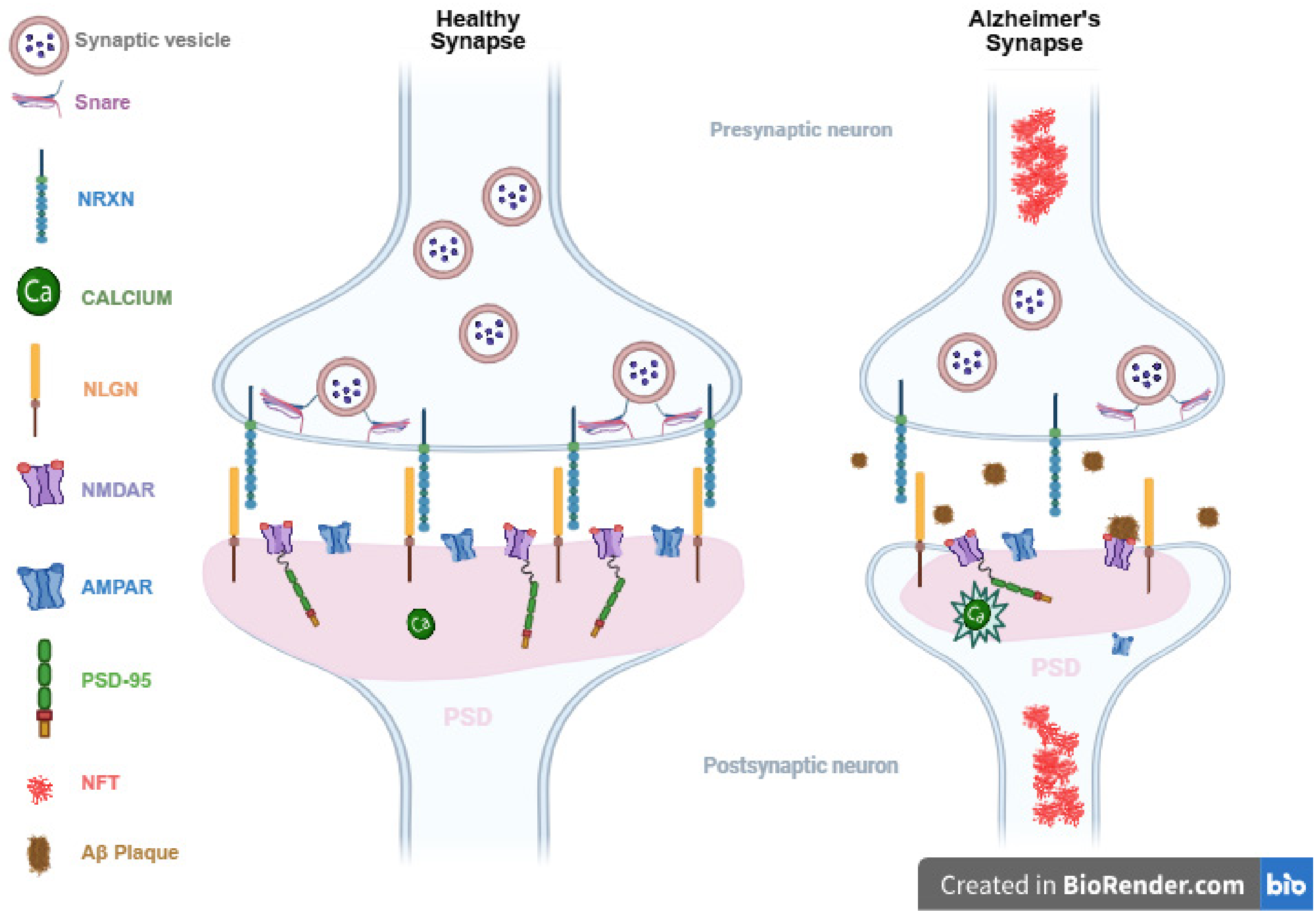

Figure 2 Neuronal transmission is enabled through the interaction between pre- and post-synaptic components. In this dynamic interaction, neurexins (NRXN) and neuroligins (NLGN) serve as pre- and post-synaptic hubs, respectively, crucial for regulating synaptic organization. Additional critical elements of this process include presynaptic organization, such as the clustering of synaptic vesicles, and postsynaptic organization, encompassing the incorporation of neurotransmitter receptors (e.g., α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptors (AMPARs) and N-methyl-D-aspartate receptors (NMDARs)) and scaffold proteins (e.g., Postsynaptic Density Protein 95 (PSD-95)). Under normal physiological conditions, synaptic architecture remains intact, maintaining a delicate equilibrium between pre- and post-synaptic components. However, in pathological states, the presence of both amyloid-beta (Aβ) and neurofibrillary tangles (NFTs) result in a reduction of both pre- and post-synaptic elements. This reduction is particularly prominent in the postsynaptic density (PSD) region, which undergoes diminishment in Alzheimer's disease (AD). Aβ plaques induce neurite dystrophy and physical obstruction, binding to NRXNs and consequently reducing NRXN surface expression. This interaction is implicated in the impairment of presynaptic organization mediated by neuroligins (NLGNs). Furthermore, Aβ plaques physically obstruct NMDAR, thereby disrupting signal transmission. Aβ oligomers and tau pathology further dysregulate calcium (Ca^{2+}) signaling, through mechanisms such as NMDARs overactivation and intracellular Ca^{2+} overload, ultimately impairing synaptic plasticity and contributing to excitotoxicity in AD. Pathogenic tau, found in NFTs, may disrupt normal synaptic vesicle release at the presynapse. Additionally, given tau's involvement in AMPAR internalization, it is conceivable that pathogenic tau could disrupt AMPAR organization.

The literature includes studies that partially elucidate the molecular mechanisms and protein-level changes underlying synaptic dysfunction in AD. The loss of synaptic connectivity may result from alterations in pre- or post-synaptic proteins, which are located in synaptic vesicles, the cytoplasm, and terminal membranes. Post-mortem analyses of human brain samples from AD patients have reported a reduction in synaptic protein levels (Figure 2) (71).

While the majority of studies have traditionally emphasized postsynaptic receptors and dendritic processes in AD, there is a growing body of evidence indicating the involvement of presynaptic processes and disruptions in presynaptic plasticity in the pathogenesis of AD. Presynaptic dysfunction is likely to exert a considerable impact on brain circuit activity and may contribute to the observed network hyperactivity in rodent models of AD as well as in AD patients (72), owing to its disruption of the delicate balance between excitatory and inhibitory synaptic transmission (41). Pathological alterations in cortical synapses have been found to correlate more closely with the severity of dementia than other hallmark pathological features of AD, such as Aβ plaques and NFTs.

Presynaptic Proteins and AD

The SNARE proteins play a fundamental role in regulating neurotransmitter exocytosis at the presynaptic site (73). Perturbations in the expression levels of SNARE proteins, including syntaxin-1, synaptosomal associated protein 25 (SNAP-25), and vesicle-associated membrane protein (VAMP), have been demonstrated to affect synaptic function in animal models, thus offering potential targets for neuropsychiatric therapeutic interventions (74). These proteins, constituting the SNARE complex, are either localized within synaptic vesicles or associated with the presynaptic plasma membrane (SNAP-25 and syntaxin). Brinkmalm et al. reported significantly reduced intensities of most evaluated tryptic peptides corresponding to SNAP-25B, syntaxin-1B, syntaxin-1A, or VAMP-2 in the AD group compared to the control group (75).

N-cadherin, a synaptic adhesion molecule, has been implicated in AD, with reports indicating elevated levels of cleaved N-cadherin in homogenates of postmortem brain tissue from AD patients compared to non-AD individuals (76). Furthermore, in individuals with AD, cerebrospinal fluid analysis revealed higher levels of cleaved N-cadherin compared to healthy controls. Notably, treatment with Aβ was associated with a reduction in N-cadherin expression in neuronal-like cells (77). These findings suggest a potential link between aberrant N-cadherin levels and the pathogenesis of AD.

Presynaptic adhesion molecules play a crucial role in specifying neuronal synapses, with notable examples including Neurexins (NRXNs), including NRXN1, NRXN2, and NRXN3. AβO deposits within damaged synapses interact with neurexins and postsynaptic neuroligins. It has been reported that NRX1, NRX2α, and NRX1β bind to AβOs (78). While the interaction of AβOs with NRX1β does not impede its ability to bind to synaptic partners such as NLGN1 or LRRTM2, AβO treatment reduces the surface expression of NRX1β (79). Notably, NRXN3 gene variants are considered to be associated with AD. NRXN3 interacts with the APOE4 haplotype and modulates the expression of NRXN3's transmembrane or soluble isoforms in postmortem AD brains. It has been proposed that dysregulation of presynaptic NRXN3 expression and splicing may exacerbate neuronal inflammation in the AD brain. Such dysregulation could represent an early event leading to disturbances in synaptic calcium homeostasis and the infiltration of AβOs at synapses (80).

A recent meta-analysis of 22 studies investigating presynaptic protein alterations in AD reported a significant decrease in presynaptic protein levels in postmortem tissue of AD patients compared to healthy controls (81).

Postsynaptic Proteins and AD

Neuroligins (NLGNs) are postsynaptic adhesion proteins that engage in interactions with presynaptic protein neurexins (NRXNs) and play pivotal roles in synapse formation, maturation, maintenance, and plasticity. All neuroligin isoforms are characterized by their transmembrane nature, residing within the postsynaptic membrane. Specifically, Neuroligin-1 (NLGN1) predominantly localizes at excitatory postsynaptic densities, where it establishes connections with key synaptic proteins such as NMDAR, PSD-95, and Synaptic Scaffolding Molecule (S-SCAM) (82).

Over the past decade, there has been increasing recognition of the significance of NLGN1 in the context of AD neurodegeneration. Studies in rats have demonstrated the deleterious impact of Aβ1-40 fibrils on NLGN1, resulting in synaptic dysfunction and memory impairment (83). Moreover, in vitro experiments have shown that AβOs bind to NLGN1, and disruption of this interaction adversely affects synaptic integrity (78). Additionally, NLGN1 levels have been observed to be reduced in the plasma of individuals with AD (84), while hippocampal NLGN1 expression was found to be decreased in AD patients compared to control subjects (85).

PSD-95 is the primary scaffold protein within the PSD of dendritic spines. The PSD has various functions, including facilitating membrane receptor anchorage in the spines, modulating trafficking and localization of adhesion molecules, ion channels, and glutamate receptors (AMPAR, mGluR, NMDAR), and regulating synaptic response and synaptic plasticity (86).

PSD-95 has been implicated in AD, aging, and various mental illnesses. However, there is inconsistency in the literature regarding PSD-95 expression in AD. Some studies indicate reduced PSD-95 expression in transgenic mouse models of aging and AD (87), while others also report decreased PSD-95 levels in brain tissue from AD mouse models (88) and individuals with AD (89). In contrast, others have observed a significant increase in PSD-95 expression, which is positively correlated with β-amyloid and phosphorylated Tau proteins in AD cases (90). Compensatory mechanisms during the progression of AD may contribute to the discrepancy observed in PSD-95 expression.

Homer scaffolding proteins are crucial components of the postsynaptic density (PSD) that facilitate the organization of multiprotein complexes involved in intracellular signaling. They interact with various PSD proteins, including receptors and ion channels, thereby influencing synaptic architecture, intracellular calcium signaling, and neuronal development (91). Homer proteins, particularly Homer1b, have been implicated in AD pathogenesis.

Evidence from research suggests that exposure to soluble Aβ leads to a significant decrease in the size of synaptic clusters containing Homer1b, indicating a potential depletion of Homer1b induced by Aβ (92). The assembly of metabotropic glutamate receptors (mGluRs) into clusters at the cell surface hinges upon their interaction with Homer1b. Consequently, Homer 1b declustering (loss) is likely to contribute to the Aβ-induced impairment of mGluRs (93).

While Homer2 has not been explicitly linked to AD pathogenesis, research indicates its involvement in inhibiting APP processing and reducing Aβ secretion (94). Consequently, the diminished expression of Homer proteins may plausibly contribute to AD pathology.

Conclusion and Future Perspectives

Although the diagnosis of AD is traditionally characterized by the presence of extracellular Aβ plaques and intracellular NFTs, synaptic loss and dysfunction, along with eventual neuronal cell death, may manifest during the early stages of the disease. Notably, synaptic loss, rather than plaque and tangle pathology, exhibits the strongest correlation with cognitive symptoms (11). As discussed throughout this review, synaptic changes, including both dysfunction and loss, are reflected in alterations in the levels of pre- and post-synaptic proteins.

Given their pivotal roles in synaptic integrity and function, pre- and post-synaptic proteins present promising biomarkers and therapeutic targets for AD. Altered levels of synaptic proteins such as PSD-95 and SNAP-25 have been identified in brain tissue and cerebrospinal fluid (CSF) of AD patients (95). Future directions may involve targeting scaffold proteins or adhesion complexes (e.g., NRXN–NLGN, N-cadherin) to restore synaptic architecture, bridging basic research and clinical translation. These findings highlight their potential utility as early diagnostic markers, as synaptic alterations often precede overt neurodegeneration and cognitive decline.

The ability to detect synaptic dysfunction through non-invasive or minimally invasive methods could transform AD management. For instance, advancements in CSF and blood-based assays targeting specific synaptic proteins offer a pathway toward earlier and more accurate diagnoses (96). Furthermore, imaging techniques capable of visualizing synaptic protein levels in vivo, such as Synaptic Vesicle Glycoprotein 2A (SV2A) Positron Emission Tomography (PET) imaging with novel tracers (97), hold significant promise for identifying at-risk individuals and monitoring disease progression.

In addition to their biomarker potential, pre- and post-synaptic proteins represent attractive therapeutic targets. Restoring synaptic function or preventing synapse loss could mitigate cognitive decline and improve outcomes for individuals with AD.

Future studies should elucidate the mechanistic pathways connecting synaptic protein alterations with cognitive decline. Large-scale longitudinal studies are needed to validate synaptic proteins as biomarkers and establish their specificity and sensitivity for AD diagnosis and progression monitoring. Meanwhile, preclinical and clinical trials should focus on leveraging these proteins as therapeutic targets, with an emphasis on translating experimental findings into practical, patient-centered applications.

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